

Week 4 — Sampling & the CLT

Business Forecasting · LIVE deck (slides + exercises in one deck)

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Where we are

C1 Framing → C2 Describing data → **C3 Uncertainty** → C4–C5 Regression → C6 Forecasting

We enter **C3 — uncertainty**: our data is only a *sample*, so any number we compute is a *guess*.

- **This week (W4)**: the **random sample**, the **estimator** and its properties (**bias, variance, MSE**), the **sampling distribution**, and the **Central Limit Theorem**.
- Next week (W5) turns this into **confidence intervals**.

This week: review, done fast in R

- You have met these ideas before — we move quickly and lean on **simulation in R** to see them.
- The CLT especially is best *simulated*: draw many samples, watch the distribution of the mean take shape.
- The exercises appear **as slides** in this deck and the code runs right here in the browser.

Motivating example

You run a bunch of **Airbnbs** in Mexico City. Should you invest more in cleaning?

- Concretely: can you charge a higher price if your cleanliness score exceeds **4.5**?
- Get a sample of listings and compare the price of
 - those with cleanliness score below 4.5 (**dirty**)
 - and above 4.5 (**clean**)

Motivating example: the sample

We have a sample of 200 listings (**price**, cleanliness **clean**):

id	review_scores_cleanliness	price	clean
40032982	3.00	1023	Dirty
21962322	4.50	4500	Dirty
41841538	4.50	380	Dirty
624813934659858771	3.40	1350	Dirty
47030021	4.29	684	Dirty
46736994	4.29	451	Dirty

Statistical language: population & parameter

- **Population**: the entire group we want to learn about — impossible to assess directly.
 - All Airbnb listings in Mexico City; ideally the whole distribution of prices.
- **Parameter**: a number describing a characteristic of the population.
 - The mean price of clean μ_c and dirty μ_d apartments.

How close does one sample get?

In real work the parameter is unknown, so you can never check your answer. In a **simulation** you can — so we start there.

Every exercise in this deck runs on one simulated population, built in the background:

```
set.seed(1)
population <- rgamma(100000, shape = 4, rate = 0.05) # a skewed "population"
```

`mean(population)` is the **parameter**. `mean(sample(population, 30))` is one **estimate** built from 30 listings.

Before you run anything: which of those two numbers changes if you repeat it?



Exercise 1 — The parameter and one estimate

1. Compute the parameter and one estimate from $n = 30$, and print them side by side.
2. Now produce **five** estimates, each from its own fresh sample of 30.
3. Report the range they cover.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

One is fixed, the other is not

- `mean(population)` — the **parameter** — is a single fixed number. It never moves.
- `mean(sample(population, 30))` — the **estimate** — is different on every draw.

The whole of C3 is about that second line: how far does the estimate wander, and what can we still say about the truth after seeing only one draw of it?

Statistical language: sample & goal

- **Sample**: the part of the population we have data for.
 - Our 200 listings.
- **Goal**: what we want to learn about the population.
 - Is $\mu_c > \mu_d$? If so, by how much?
 - We don't know μ_c, μ_d — we **guess** them with an **estimator** and a random IID sample.

What is a random sample?

- **At random:** each member of the population has an equal chance of being selected. Selecting is called *drawing* a sample.
- **Random variable P_i :** describes observation i *before* drawing — it could be any price from the distribution.
- **Random sample:** a collection of random variables $\{P_1, P_2, \dots, P_n\}$.
- **Observed value p_i :** once realized, it is a constant from our sample — no longer random.

Before drawing the sample

The P_i are still random — no values yet:

Random Variables	P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs							
Realized Values p_i							

Random sample — Draw 1

Drawing realizes each P_i into a concrete value p_i :

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	8451	9015	8161	9085	8268	1622
Realized Values p_i	120	150	800	200	1400	110

Random sample — Draw 2

A different draw from the **same** population gives **different** realized values:

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	1933	3947	3145	3773	6721	3373
Realized Values p_i	260	420	500	2120	800	1450

The point estimate changes **every** time we draw.

IID samples

- **IID** = Independent and Identically Distributed.
- **Independent:** selecting one unit P_i doesn't affect selecting another P_j .
- **Identically distributed:** all units P_i come from the same distribution.

Does it matter *how* you pick the 30?

Two ways to end up with 30 listings:

- `sample(population, 30)` — 30 drawn at random.
- `head(sort(population), 30)` — the 30 cheapest listings, the ones a price-sorted search page hands you first.

Both produce a number. Are both of them estimates of the population mean?



Exercise 2 — Which scheme is IID?

1. Draw the random sample 1,000 times and average the 1,000 sample means.
2. Compute the mean of the 30 cheapest listings.
3. Compare both to `mean(population)`, and name which IID condition the second scheme breaks.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 mu <- mean(population)
2
3 # (1) random sampling, repeated:
4
5 # (2) the 30 cheapest listings:
```

Estimators: intuition

- Our **method of guessing** the parameter from the data.
- A function of the random variables in the sample: $\hat{\theta} = f(P_1, \dots, P_n)$.
- Because it is random, we can analyze its statistical properties.
- It **cannot** contain unknown quantities (like σ or μ).

Estimators: ones we've seen

- Sample mean: $\hat{\mu}_c = \bar{P} = \frac{\sum_n P_i}{n}$
- Sample standard deviation: $s_c = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (P_i - \bar{P})^2}$
- Each is a rule that turns a sample into a single guess.

Point estimate

- A single number computed from the **realized** data $\{p_1, \dots, p_n\}$:

$$\bar{p} = f(p_1, \dots, p_n) = \frac{\sum_n p_i}{n}$$

- No longer random — it is one number from one sample.

One number isn't enough

- The mean price of clean listings in our sample is $\bar{p}_c = 1245.4$ MXN — our point estimate.
- We **can't** say how close this one number is to the true mean without knowing the population.
- But we **can** judge our *method* (the estimator) by looking at **every estimate it could have produced**.



Exercise 3 — Once, then two thousand times

1. Draw one sample of $n = 30$ from `population`, store it in `s`, compute `mean(s)`.
2. Now do that 2,000 times. Collect the 2,000 means in `means`.
3. `hist(means)`, then mark the parameter on the picture with `abline(v = mean(population), col = "red", lwd = 3)`.

Look at the picture before moving on: where is it centred, and how wide is it compared with the prices themselves (`hist(population)`)?

Exercise

↺ Start Over

💡 Show Hint

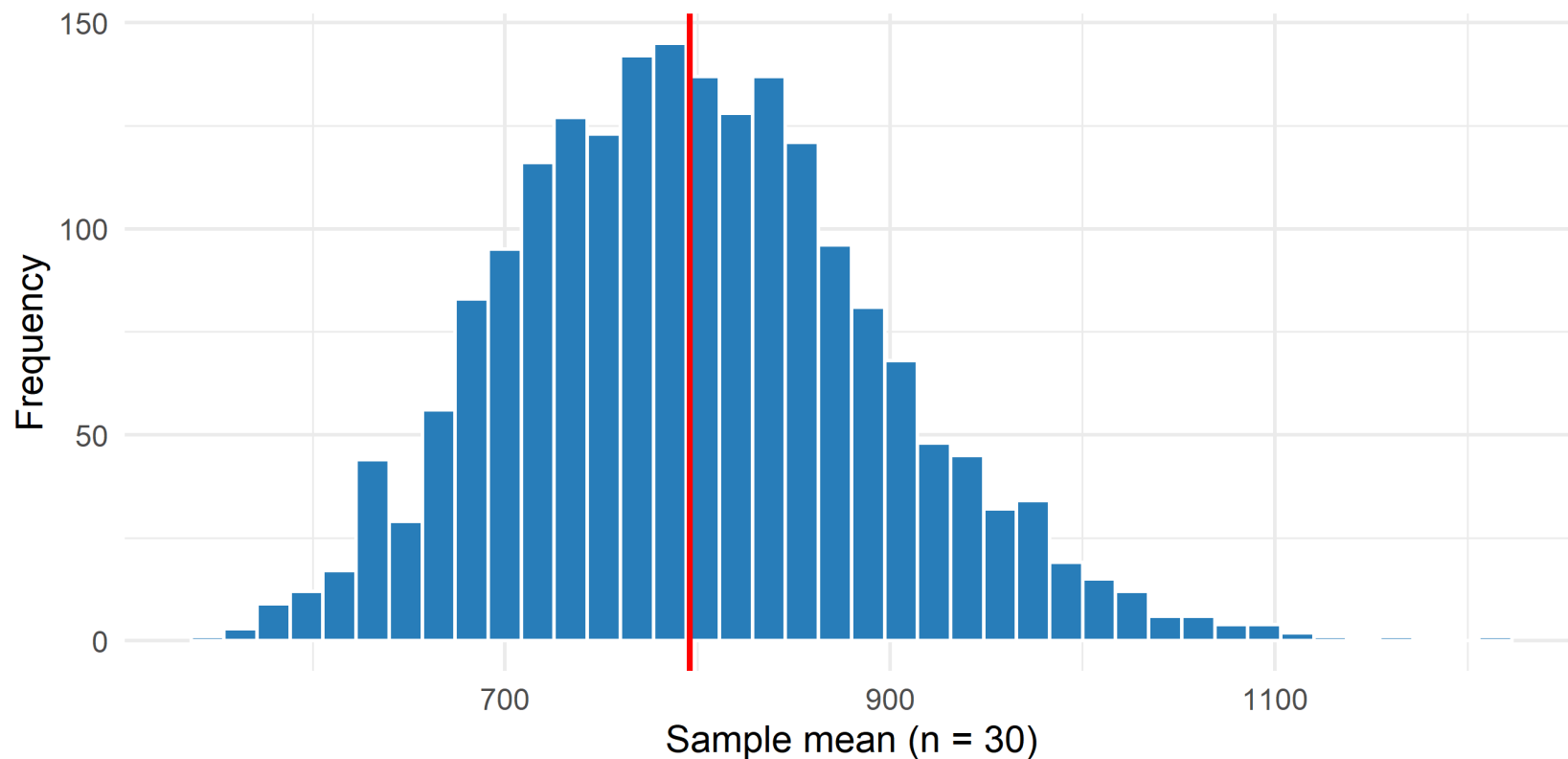
▶ Run Code

```
1 # Your code here
```

What you just built: the sampling distribution

The distribution of an estimator across all the samples it could have been computed from.

```
set.seed(1)
population <- rlnorm(100000, meanlog = 6.5, sdlog = 0.6) # skewed prices
means <- replicate(2000, mean(sample(population, 30)))
ggplot(data.frame(means), aes(means)) +
  geom_histogram(bins = 40, fill = "#2980b9", color = "white") +
  geom_vline(xintercept = mean(population), color = "red", linewidth = 1.2) +
  labs(x = "Sample mean (n = 30)", y = "Frequency") + th
```



The means cluster around the **true mean** (red) — the estimator has a *distribution*.

How wide is that histogram?

Individual prices are spread out: `sd(population)` is about 40.

The sample means sat in a visibly narrower band. **How much narrower?**

Write down a guess for `sd(means)` at $n = 30$ before you compute it.



Exercise 4 — Guess the spread, then measure it

Your guess for `sd(means)`: _____

Now rebuild `means` and compute three numbers:

1. `sd(population)` — the spread of individual prices.
2. `sd(means)` — the spread of the sample means.
3. the ratio `sd(population) / sd(means)`.

That third number is one you recognise. Which number is it?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```



Exercise 5 — Break it: change n

Here is working code. Change **one** thing — the sample size — and report what happens to the **centre** and to the **spread**.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- 5
2 means <- replicate(2000, mean(sample(population, n)))
3 c(centre = mean(means), spread = sd(means))
4
5 # Run it again with n = 30, n = 100, n = 500.
6 # Which of the two numbers moves? By how much when n goes from 5 to 500?
```



Exercise 6 — As n grows...

As the sample size n grows, the sampling distribution of the mean gets ____

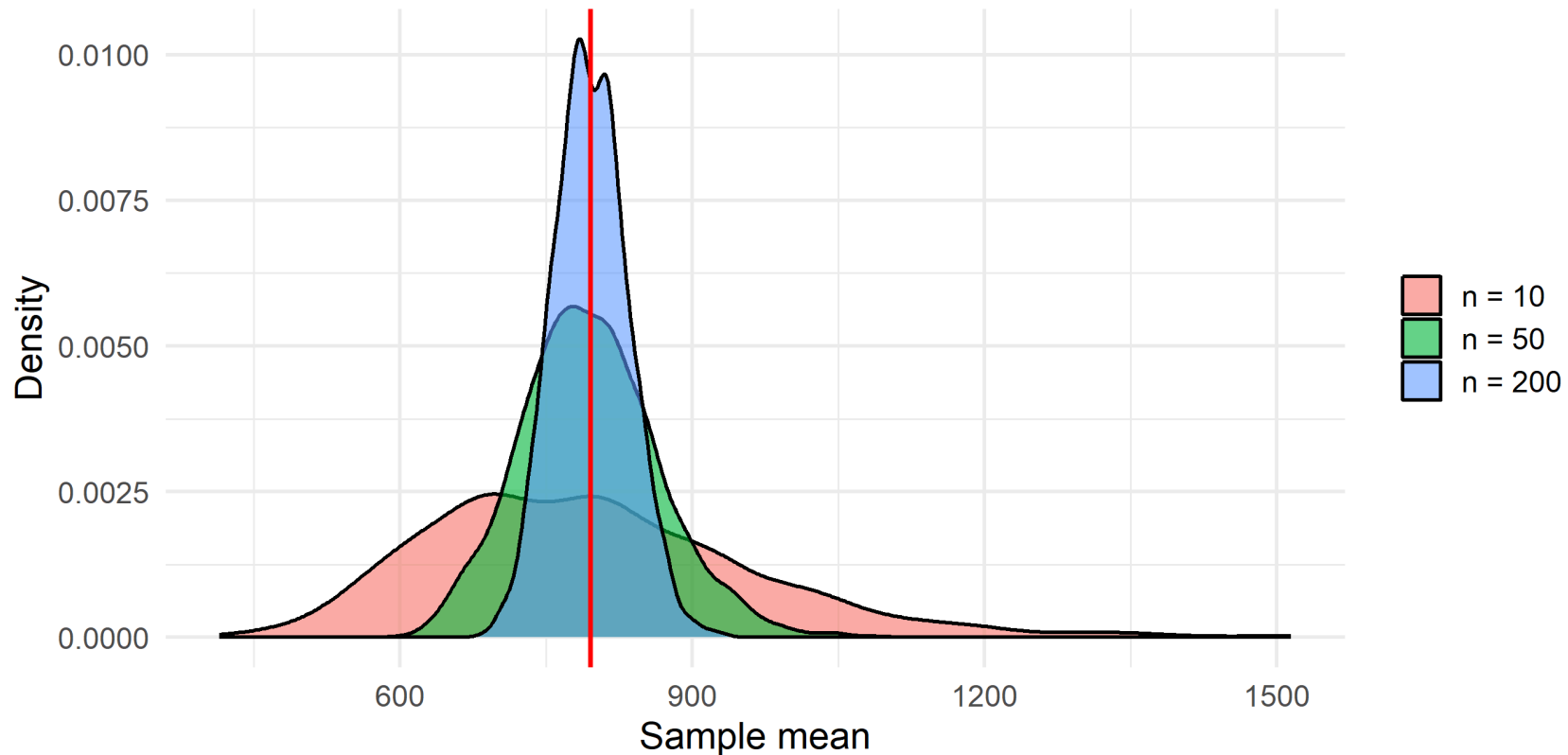
- narrower (smaller spread)?
- wider?
- more skewed?
- shifted away from the true mean?

Narrower — and by a specific amount: the spread you measured is $sd(\text{population})/\sqrt{n}$, which shrinks as n grows while the centre stays on the parameter.

The spread shrinks with $\sqrt{n}$

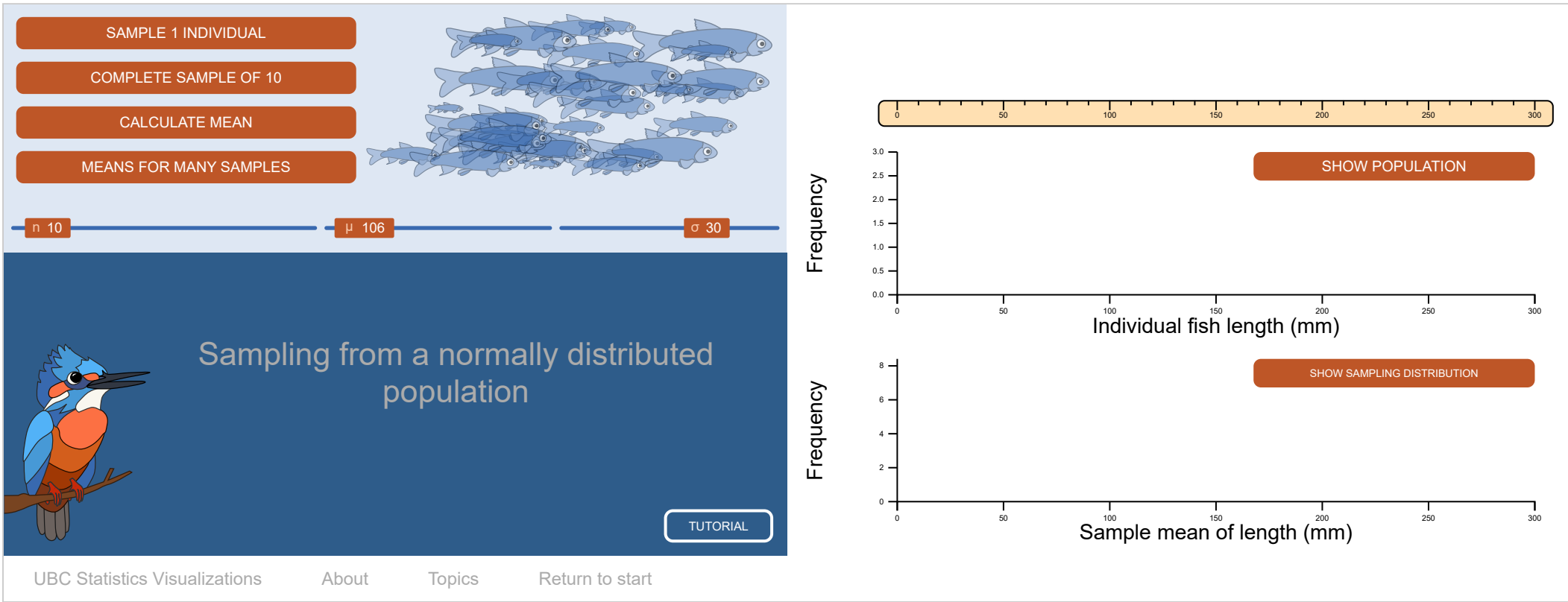
Same population, three sample sizes — centred in the same place, tighter every time:

```
set.seed(7)
pop <- rlnorm(100000, meanlog = 6.5, sdlog = 0.6)
df <- do.call(rbind, lapply(c(10, 50, 200), function(n)
  data.frame(mean = replicate(1500, mean(sample(pop, n))), n = paste0("n = ", n))))
df$n <- factor(df$n, levels = c("n = 10", "n = 50", "n = 200"))
ggplot(df, aes(mean, fill = n)) +
  geom_density(alpha = 0.6) +
  geom_vline(xintercept = mean(pop), color = "red", linewidth = 1) +
  labs(x = "Sample mean", y = "Density", fill = NULL) + th
```



Sampling distribution: explore it

Each estimator has its own sampling distribution — try a few:



From sampling distribution to the CLT

- We know the **mean** and **variance** of the sample mean. Can we say anything about its **whole distribution**?
- For the sample mean — **yes!**
- That's the **Central Limit Theorem**, the most exciting theorem in statistics.

CLT: the setup

$X_1, \dots, X_n$ are **i.i.d.** with mean μ and st. dev. σ .

- Sum $S_n = \sum_n X_i$: $E[S_n] = n\mu$, $sd(S_n) = \sqrt{n} \sigma$.
- Mean $\bar{X}_n = \frac{S_n}{n}$: $E[\bar{X}_n] = \mu$, $sd(\bar{X}_n) = \frac{\sigma}{\sqrt{n}}$.
- That $\frac{\sigma}{\sqrt{n}}$ has a name: the **standard error** of the mean — the number you measured in Exercises 4 and 5.
- Standardized $Z_n = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}}$: $E[Z_n] = 0$, $sd(Z_n) = 1$.

CLT: the theorem

For large n :

$$S_n \dot{\sim} \mathcal{N}(n\mu, \sqrt{n} \sigma), \quad \bar{X}_n \dot{\sim} \mathcal{N}\left(\mu, \frac{\sigma}{\sqrt{n}}\right), \quad Z_n \dot{\sim} \mathcal{N}(0, 1)$$

- In large samples $\bar{X}_n$ is **approximately normal**, centered at μ , with st. dev. $\frac{\sigma}{\sqrt{n}}$.
- As n grows, the approximation improves and the spread **shrinks with** $\sqrt{n}$.



Exercise 7 — Theory vs. simulation

The CLT claims $\bar{X}_{30} \sim \mathcal{N}(\mu, \sigma/\sqrt{30})$. Use it, then check it.

1. **Theory, no data:** with `mu <- mean(population)` and `se <- sd(population)/sqrt(30)`, compute the probability that a sample of 30 returns a mean above **90**.
2. **Simulation:** build 5,000 sample means and compute the share of them above 90.
3. How close are the two answers?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

1 # Your code here

Does the shape of the population matter?

`population` is skewed, and its sample means still came out symmetric. Maybe a gamma is a lucky case.

Two more populations are loaded:

- `pop_bimodal` — two clumps of listings: cheap rooms around 40, design apartments around 220.
- `pop_binary` — a 0/1 variable: is the listing a superhost?

Neither is remotely bell-shaped. Guess what the histogram of **their** sample means looks like.



Exercise 8 — Start from something not normal

Put three pictures side by side with `par(mfrow = c(1, 3))`:

1. `hist(pop_bimodal)` — the population itself.
2. the histogram of 2,000 sample means with `n = 5`.
3. the same with `n = 50`.

Then repeat the whole thing starting from `pop_binary`.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

CLT: what drives the shape

- The **original** distribution of X_i doesn't matter (but outliers slow convergence).
- Larger $n \rightarrow$ tighter distribution around the mean.
- Smaller $\sigma \rightarrow$ tighter distribution around the mean.

This site is archived for reference

English



Chapter 3: Probability Distributions

Probability Distribution

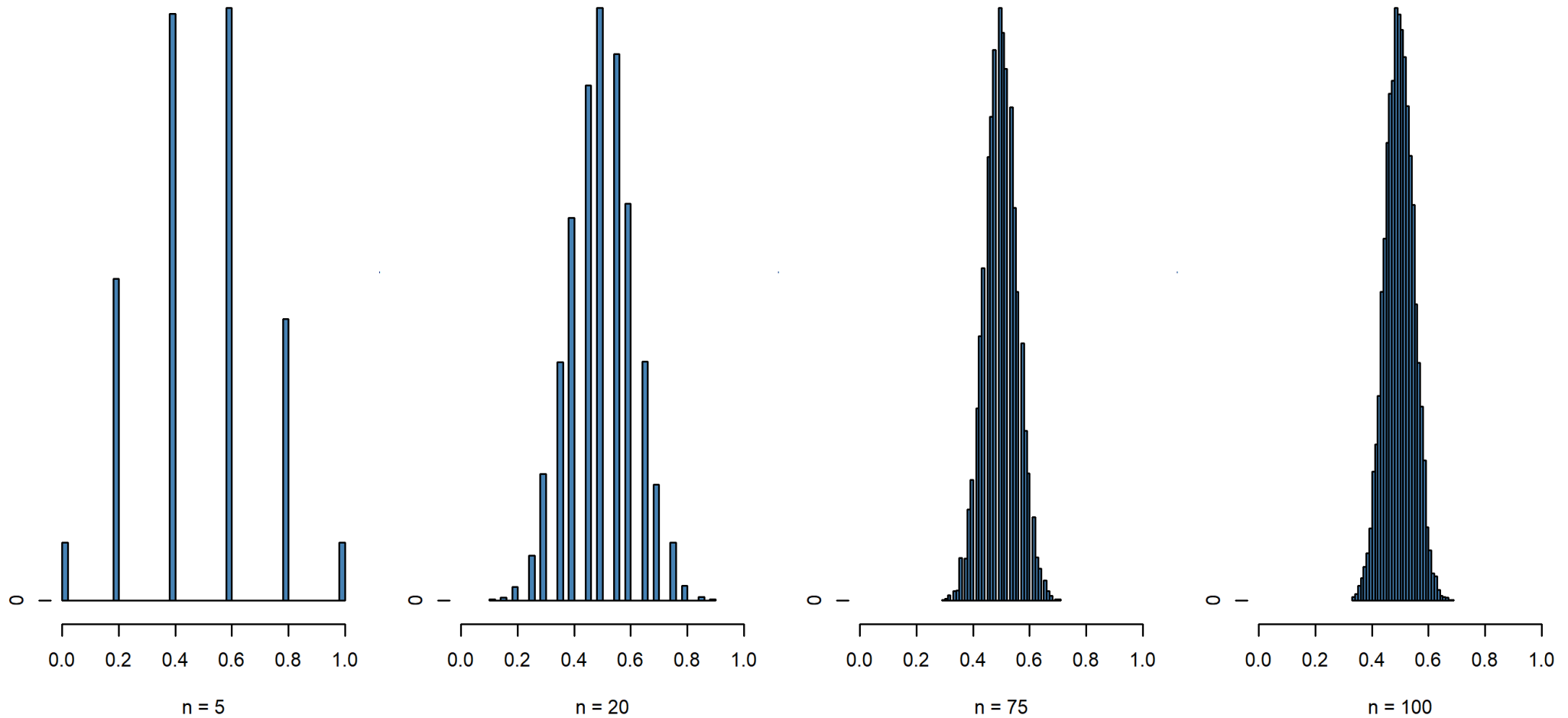
A probability distribution specifies the relative likelihoods of all possible outcomes.



Random Variable

CLT: simulate it (even for discrete data)

$X_i \sim \text{Bernoulli}(0.5)$. Distribution of $\bar{X}_n$ for growing n — it becomes normal and **tighter**:



- St. dev. of the mean: $\sigma_{\bar{X}} = \frac{\sigma_X}{\sqrt{n}} = \frac{\sqrt{p(1-p)}}{\sqrt{n}} = \frac{0.5}{\sqrt{n}}$.



Exercise 9 — Break it: more replications

A classmate says: “*my standard error is too big — I’ll just run more replications.*”

Test the claim. Hold `n = 30` fixed and build `means` with **100**, **1,000** and **10,000** replications. Report `sd(means)` each time, and say what *does* change.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # Your code here
```

Normal distribution: combining variables

A customer who opens the DiDi app may call a car. X = calls in Cancún (Bernoulli 0.4), Y = in Puerto Vallarta (Bernoulli 0.6). Sample: 100 from Cancún, 80 from PV. **P(more than 100 people call a car)?**

Reminders (for normal X):

$$X + c \sim \mathcal{N}(\mu + c, \sigma), \quad cX \sim \mathcal{N}(c\mu, |c|\sigma)$$

$$X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sqrt{\sigma_1^2 + \sigma_2^2})$$

What if I don't know σ ?

Store sales are normal with mean 200 and **unknown** variance. Take $n = 80$ stores: $P(\bar{X} > 220)$?

- By the CLT, $\bar{X} \sim \mathcal{N}\left(200, \frac{\sigma}{\sqrt{80}}\right)$ — but σ is unknown.
- Use the **sample** standard deviation to estimate σ .
- That adds a little uncertainty, but in a large sample you estimate σ well.
- The distribution still converges to normal — you just need a few more observations.



Exercise 10 — The whole week, one question

Back to the Airbnb portfolio. Your investor asks:

“You surveyed 25 listings and got an average nightly price. How wrong could that number be?”

Answer with a **number**, a **picture**, and **one sentence**.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 # 1. A number: how far from the truth does an estimate from n = 25 typically land?
2
3 # 2. A picture: the sampling distribution for n = 25, with the truth marked
4 #    and the range that holds 90% of the estimates.
5
6 # 3. One sentence: what do you tell the investor?
```

Exam question — Fall 2025 Midterm 1

The sampling/estimation logic here is the **foundation** for the Midterm's inference questions.

Last year's Midterm 1 CI questions (Q7 / Q8) build directly on the **sampling distribution of the mean** and $SE = \sigma / \sqrt{n}$ — but they ask for **confidence intervals**, which we build in **Week 5**. This week, make sure the *estimator + sampling distribution* logic is second nature. **We solve the full CI questions together next week.**

→ [More exam practice at the end of this deck](#)

What you must be able to do this week

- Tell a **random variable** P_i from a **realized value** p_i , and state what makes a sample IID.
- Write an **estimator** as a function of the sample and compute a **point estimate**.
- Check an estimator for **bias**, compare two by **variance/efficiency**, and combine both via **MSE = Var + Bias²**.
- Describe the **sampling distribution** of the mean and **simulate** it in R.
- State the **CLT**: $\bar{X}_n \approx \mathcal{N}(\mu, \sigma/\sqrt{n})$ — approximately normal, centered at μ , spread **shrinking with** $\sqrt{n}$.

Next week (W5): turn the sampling distribution into **confidence intervals**.

Appendix — optional theory refresher

Example: the estimator (before drawing)

We want the average price in Mexico City but lack the whole population. Take a sample of 8 listings and average.

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs						
Realized Values p_i						

Estimator: $\hat{\mu} = \frac{P_1 + \dots + P_8}{8}$

Example: the estimator (Sample 1)

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	8451	9015	8161	9085	8268	1622
Realized Values p_i	120	150	800	200	1400	110

Point estimate (all 8): $\frac{p_1 + \dots + p_8}{8} = 685$

Example: the estimator (Sample 2)

Random Variables P_i	P_1	P_2	P_3	P_4	P_5	P_6
Selected Listing IDs	1933	3947	3145	3773	6721	3373
Realized Values p_i	260	420	500	2120	800	1450

Point estimate (all 8): $\frac{p_1 + \dots + p_8}{8} = 809.9$ — a different sample, a different estimate.

Expectation of an estimator: bias

- A good estimator is **unbiased**: $E[\hat{\theta}] = \theta$ for any value of θ .
- Its sampling distribution is **centered** at the parameter — on average it hits the truth.

$$Bias(\hat{\theta}) = E[\hat{\theta}] - \theta$$

- Take many samples, compute the estimator in each, and average those values: does the average land on the parameter?

Three candidate estimators

Same job: guess μ from an IID sample.

- $\hat{\theta}_1 = X_1$ — just the first observation.
- $\hat{\theta}_2 = \frac{3X_1 + X_2}{5}$
- $\hat{\theta}_3 = \bar{X} = \frac{\sum_n X_i}{n}$

Which of them are **unbiased**? And is unbiased the same as good?



Exercise 11 — Which ones are unbiased?

Draw 5,000 samples of $n = 10$ from `population`. In each sample compute all three estimators. Then compare the **average** of each to `mean(population)` — that difference is the **bias**. Also record how spread out each estimator is.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 mu <- mean(population)
2 # one draw: x <- sample(population, 10)
3 # build all three estimators from x, then repeat 5,000 times
```

Example 1: $\hat{\theta}_1 = X_i$ (a single observation)

- X_i has unknown mean $E(X_i) = \mu$; we want to estimate μ .
- Estimator: $\hat{\theta}_1 = X_i$.
- Expected value: $E(\hat{\theta}_1) = E(X_i) = \mu$.
- Bias: $E(\hat{\theta}_1) - \mu = 0 \rightarrow$ **unbiased**.
- Is it a *good* estimator? (We'll come back to this.)

Example 2: $\hat{\theta}_2 = (3X_1 + X_2)/5$

- Same setup: $E(X_i) = \mu$.
- Expected value: $E(\hat{\theta}_2) = \frac{3}{5}\mu + \frac{1}{5}\mu = \frac{4}{5}\mu$.
- Bias: $\frac{4}{5}\mu - \mu = -\frac{1}{5}\mu \rightarrow$ **biased** (weights don't sum to 1).

Example 3: $\hat{\theta}_3 = \bar{X} = \frac{\sum X_i}{n}$ (sample mean)

- Same setup: $E(X_i) = \mu$.
- Expected value: $E(\bar{X}) = \frac{\sum_n E(X_i)}{n} = \frac{n\mu}{n} = \mu$.
- Bias: 0 → **unbiased**. The sample mean is our workhorse estimator.

Two unbiased estimators — is one of them better?

Sample IID from $X \sim \mathcal{N}(10, 10)$ and take four observations. Two ways to combine them:

- A: $\hat{\mu}_A = \frac{3X_1 + X_2}{4}$
- B: $\hat{\mu}_B = \frac{X_1 + X_2 + X_3 + X_4}{4}$

The weights sum to 1 in both, so both are **unbiased** — bias cannot separate them.

Which one would you rather report, and what evidence would settle it?

Exercise 12 — Which unbiased estimator wins?

Simulate 1,000 samples of four draws from $\mathcal{N}(10, 10)$, compute both estimators in each sample, then produce the evidence: their averages, their standard deviations, and the two histograms side by side.

Exercise

 Start Over

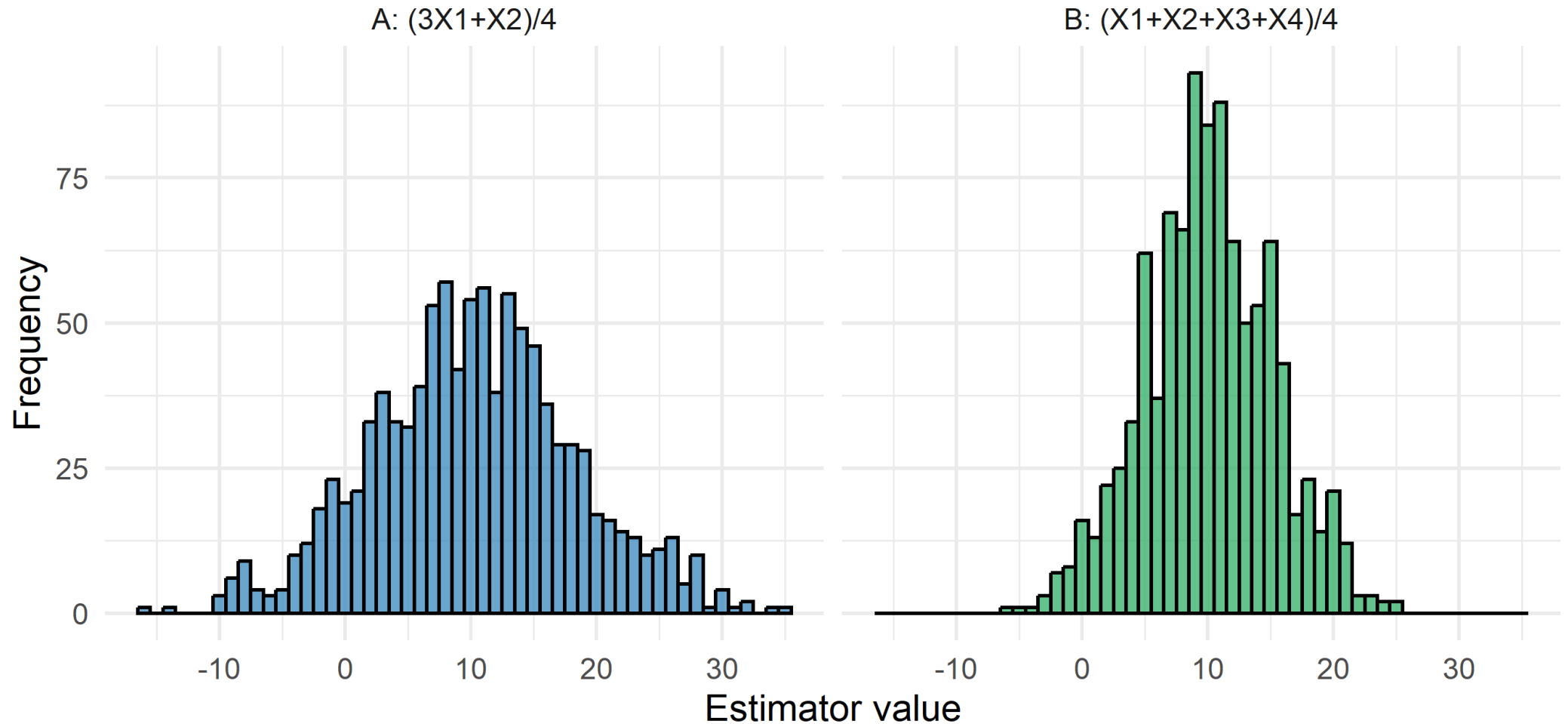
 Show Hint

 Run Code

```
1 # one sample of four draws:  x <- rnorm(4, mean = 10, sd = 10)
2 # build A and B from x, repeat 1,000 times, then compare and plot
```

Variance of an estimator: efficiency

Among **unbiased** estimators, prefer the one with **smaller variance** (more *efficient*).



B is tighter — more efficient.

Variance: definition & relative efficiency

$$Var(\hat{\theta}) = E[(\hat{\theta} - E[\hat{\theta}])^2]$$

- Lower variance = more efficient (a property of *unbiased* estimators). Here:

$$Var(\hat{\mu}_A) = \frac{10\sigma^2}{16} > \frac{4\sigma^2}{16} = Var(\hat{\mu}_B)$$

- Relative efficiency: $Eff(\hat{\mu}_B, \hat{\mu}_A) = \frac{Var(\hat{\mu}_A)}{Var(\hat{\mu}_B)} = \frac{10}{4} = 2.5$ — B is 2.5× as efficient.

Variance: worked examples

- $\hat{\theta}_1 = X_i: Var(\hat{\theta}_1) = \sigma^2$
- $\hat{\theta}_3 = \bar{X} = \frac{\sum X_i}{n}: Var(\hat{\theta}_3) = \frac{\sigma^2}{n} - \text{shrinks with } n.$
- $\hat{\mu}_A = \frac{3X_1 + X_2}{4}: Var(\hat{\mu}_A) = \frac{9\sigma^2 + \sigma^2}{16} = \frac{10\sigma^2}{16}$

Side note: what if variables aren't independent?

So far we assumed independence. Suppose X_1, X_2 are **not** independent (e.g. daily sales of two products in one store).

- $E(X_1 + X_2) = E(X_1) + E(X_2)$ (always true).
- $Var(X_1 + X_2) = Var(X_1) + Var(X_2) + 2 Cov(X_1, X_2)$
- $Var(X_1 - X_2) = Var(X_1) + Var(X_2) - 2 Cov(X_1, X_2)$

Mean Squared Error (MSE)

A single summary of how good an estimator is:

$$MSE(\hat{\theta}) = E[(\hat{\theta} - \theta)^2]$$

- The **lower** the MSE, the better. It combines **bias** and **variance**.

MSE = Variance + Bias²

$$\begin{aligned}MSE(\hat{\theta}) &= E[(\hat{\theta} - \theta)^2] = E[(\hat{\theta} - E(\hat{\theta}) + E(\hat{\theta}) - \theta)^2] \\&= E[(\hat{\theta} - E(\hat{\theta}))^2] + 2 \underbrace{(E[\hat{\theta} - E(\hat{\theta})])}_{=0} (E(\hat{\theta}) - \theta) + (E(\hat{\theta}) - \theta)^2 \\&= \text{Var}(\hat{\theta}) + \text{Bias}(\hat{\theta})^2\end{aligned}$$

- If unbiased, $MSE(\hat{\theta}) = \text{Var}(\hat{\theta})$.

Trading bias for variance

Estimating customer income; the true value is $\theta = 10$. Two candidates:

- **A** — biased by -0.2 , but almost no noise (sd 0.1)
- **B** — unbiased, but noisy (sd 1)

Unbiasedness alone says take B. Which one would you actually report? Produce the numbers and the pictures before you decide.



Exercise 13 — Put a number on the trade-off

Simulate 5,000 draws of each estimator. Then, for both:

1. Compute the **bias**, the **variance**, and the **MSE** two ways — straight from the definition `mean((est - theta)^2)`, and from the decomposition $Var + Bias^2$.
2. Draw the two sampling distributions side by side on a common x-axis, with the true value marked.
3. Say which one you report, and why.

Exercise

↺ Start Over

💡 Show Hint

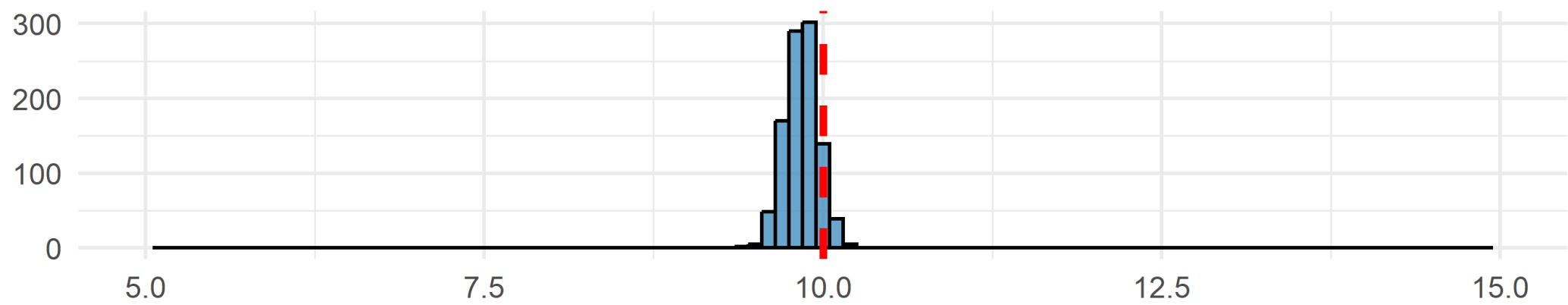
▶ Run Code

```
1 theta <- 10
2 A <- rnorm(5000, mean = 9.8, sd = 0.1)
3 B <- rnorm(5000, mean = 10, sd = 1)
4
5 # bias, variance, MSE (both routes) for each; then the two histograms
```

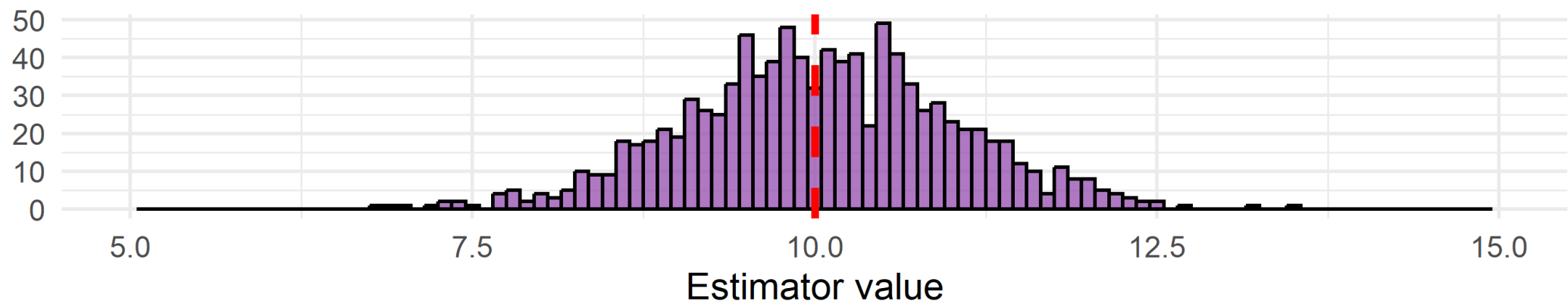
Trading bias for variance: the picture

Red dashed line = true value.

Biased, low variance



Unbiased, high variance



Sampling distribution of the standard deviation

- Sample **means** are normal in large samples. What about the sample **standard deviation**?
- If we take many samples and compute each one's SD, what does *that* distribution look like?
- If X_i is normal, the sample variance is related to the **chi-squared** distribution.
- This is what we'll use to build confidence intervals for the variance.

From Normal to Chi-Square

Start with the standard normal $Z \sim N(0, 1)$ and **square it**. The distribution of $X = Z^2$ is called **chi-squared with one degree of freedom**, $\chi^2(1)$.

That is the definition. The rest you can find out yourself:

- What shape does it have, and what values can it take?
- What are its mean and its variance?
- If $X = Z^2$, how does $P(X > 1)$ relate to a probability about Z ?



Exercise 14 — Square a normal

1. Draw 100,000 standard normals into `z` and square them into `x`.
2. Plot the distribution of `x`, and compute `mean(x)` and `var(x)`.
3. Compute `mean(x > 1)` and `mean(abs(z) > 1)`. Are they equal? Say in one line why.

Exercise

↺ Start Over

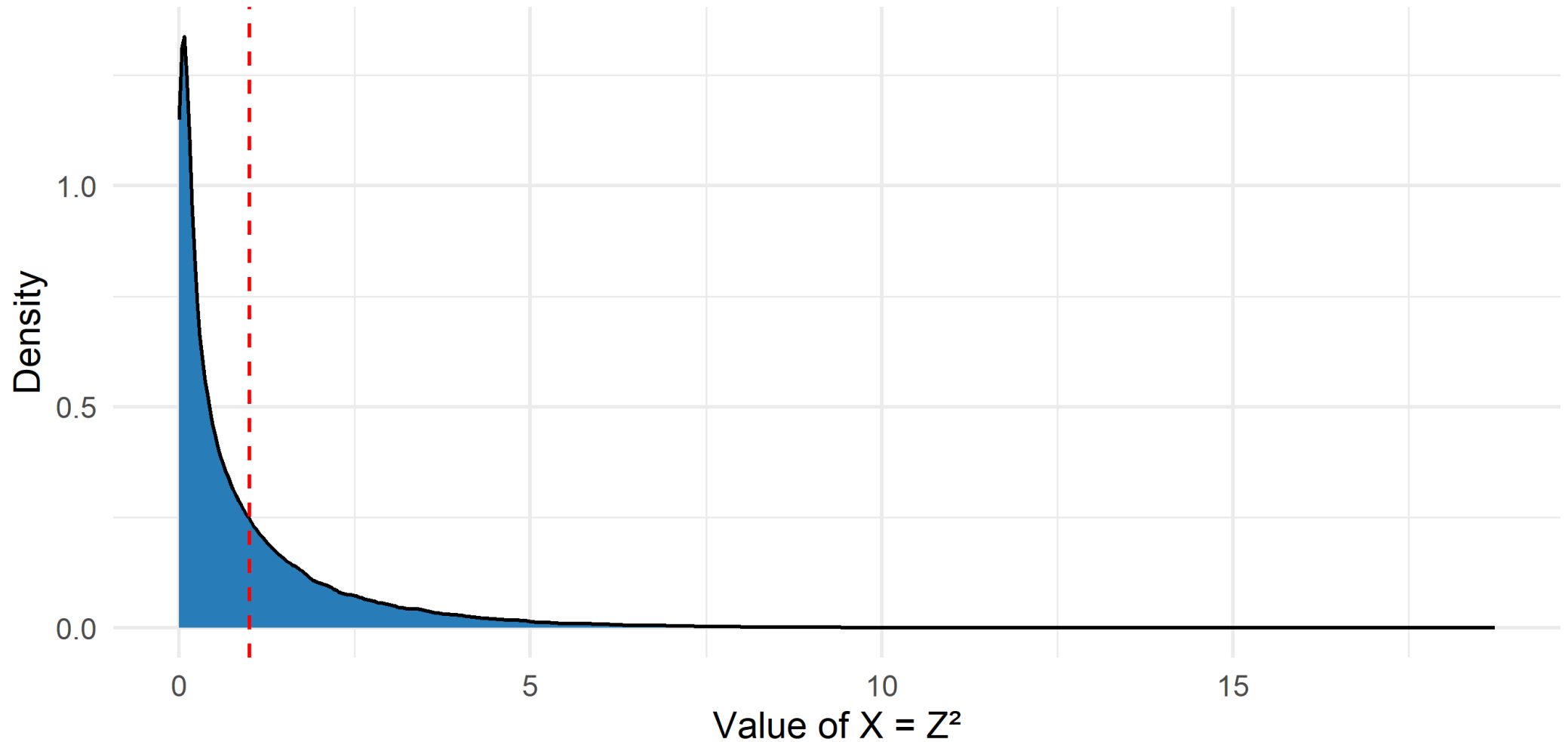
💡 Show Hint

▶ Run Code

```
1 # Your code here
```

Chi-Square: the picture

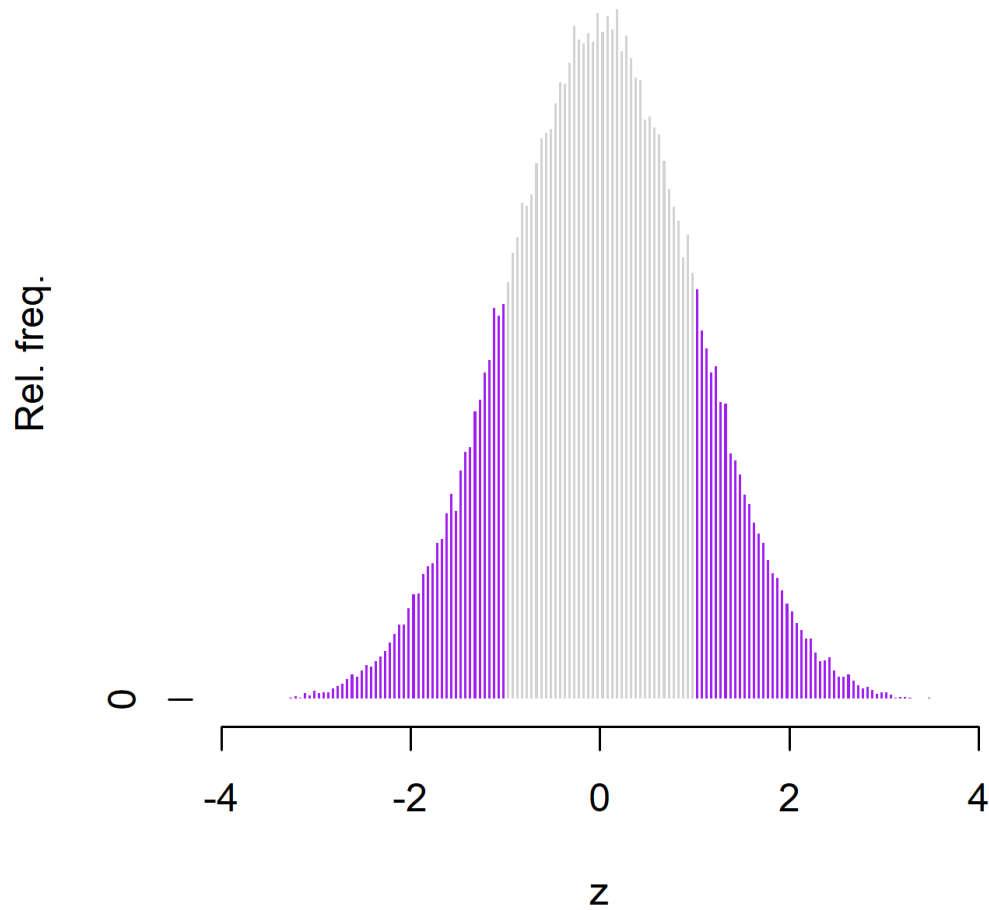
- $E[\chi^2(1)] = \text{Var}(Z) = 1; \text{Var}(\chi^2(1)) = 2.$



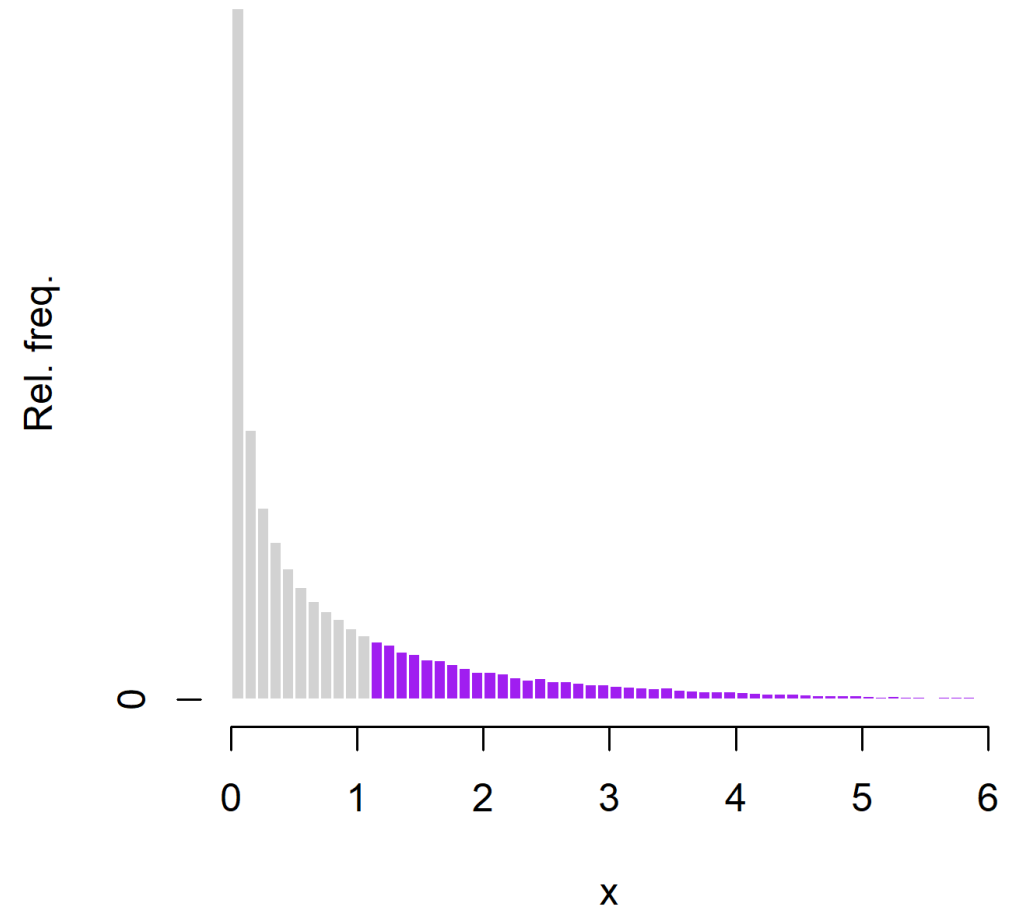
Visualizing the connection

Shaded area = $P(X = Z^2 > 1)$ where $X \sim \chi^2(1)$, $Z \sim N(0, 1)$ — same area in both panels:

Standard Normal

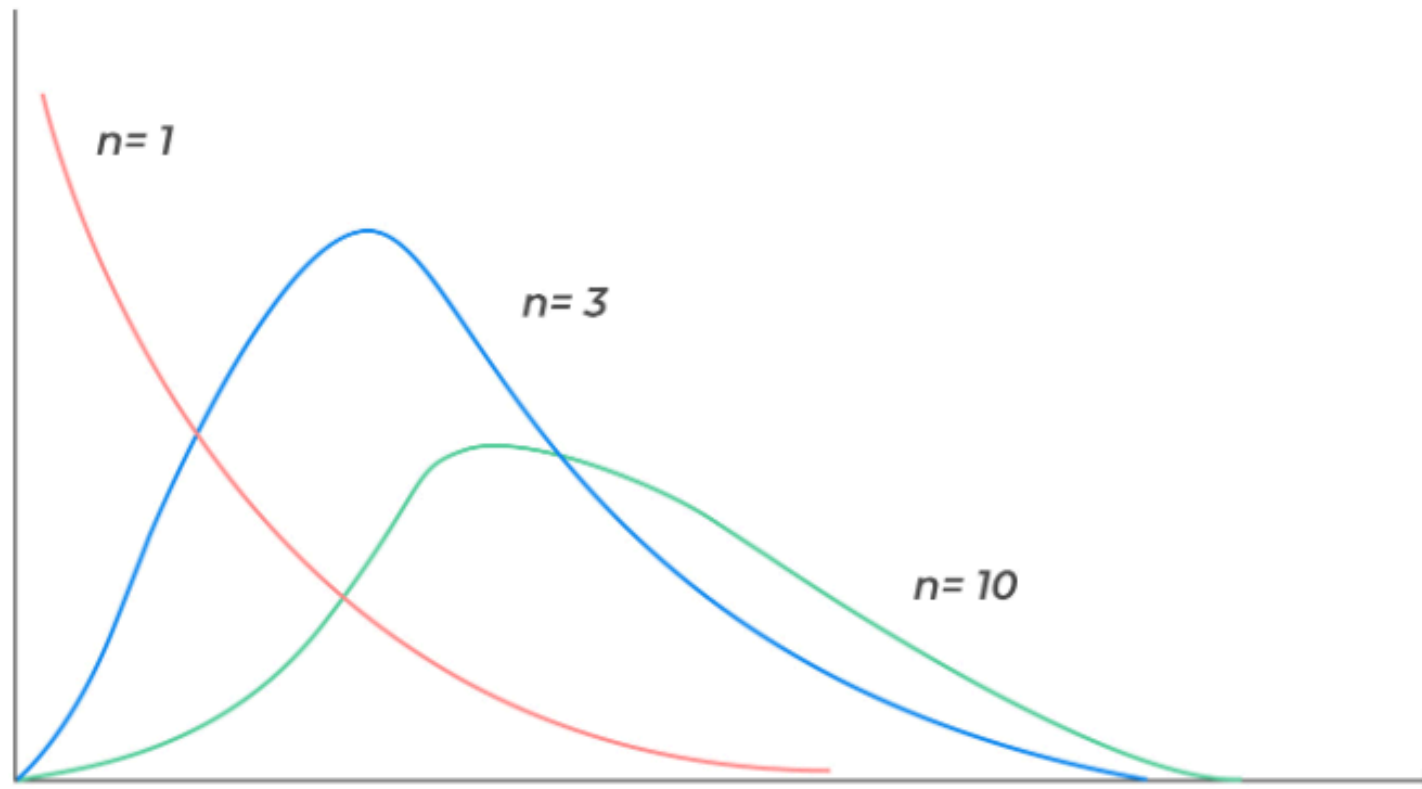


Chi-Squared(1)



Chi-Square & the sum of normals

- Sum of n i.i.d. squared standard normals is $\chi^2(n)$: $\sum_n Z^2 \sim \chi^2(n)$.
- $E[\chi^2(n)] = n$; $Var(\chi^2(n)) = 2n$.



- Why does the shape approach normal for large n ? Because it's a **sum of random variables** — the CLT again.

Exam practice

These problems mirror the **weekly exercise set** and the style of the **exam** — work each one in the code cell, then press **Solution** to reveal the full worked answer.

[← Back to the exam spotlight](#)



Exam practice 1 — Liga MX winning percentages

Liga MX teams play **34 games** per season. A team's final winning percentage is its number of wins divided by 34. Assume each match is an i.i.d. Bernoulli trial with the team's "true" win probability. [11]

- If Club América has win probability $\pi_A = 0.58$, what is the asymptotic distribution of América's winning percentage W_A ?
- If América has $\pi_A = 0.58$ and Pumas has $\pi_B = 0.52$, what is the asymptotic distribution of $W_A - W_B$, assuming independence?
- Based on (b), what is $P(W_B > W_A)$?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- 34
2 piA <- 0.58; piB <- 0.52
3
4 # (a) the mean and the variance of America's winning percentage
5 # (b) the mean and the variance of the difference W_A - W_B
6 # (c) the probability that that difference comes out negative
```



Exam practice 2 — Factory injuries (Poisson)

In a factory in Monterrey, the number of workplace injuries is recorded weekly for **240 weeks**. The average number of injuries per week is **0.6**, with $X_t \sim \text{Poisson}(\lambda)$. [12]

a. Provide an asymptotic 90% confidence interval for λ .

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 lambda_hat <- 0.6
2 n          <- 240
3
4 # (a) the standard error of lambda-hat, then the 90% interval around it
```



Exam practice 3 — Unemployment duration: concepts

A random sample of **500** unemployed workers in Guadalajara is surveyed about unemployment duration in weeks, $\bar{X}$. By the CLT, $\bar{X}$ is approximately normal. [18]

- a. If $P(\bar{X} > 20) = 0.5$, what is $E(X)$?
- b. If $P(\bar{X} > 20) = 0.4$, what can be said about $E(X)$?

Answer them by finding the mean that produces each probability. Take $\sigma_X = 10.5$, so $SE = 10.5/\sqrt{500}$.

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 se <- 10.5 / sqrt(500)
2
3 # (a) the mean that makes P(xbar > 20) equal to 0.50
4 # (b) the mean that makes P(xbar > 20) equal to 0.40
```

Exam practice 3 — Unemployment duration: calculations

Same setup: $n = 500$ workers, $\bar{X}$ approximately normal by the CLT. [18]

c. If $E(X) = 19.5$ and $\sigma_X = 10.5$, what is $P(\bar{X} > 20)$?

d. How would the answer to (c) change if $n = 1600$ instead of $n = 500$?

Exercise

 Start Over

 Show Hint

 Run Code

```
1 mu    <- 19.5
2 sigma <- 10.5
3
4 # (c) the probability that the sample mean exceeds 20 when n = 500
5 # (d) the same probability when n = 1600
```



Exam practice 4 — Cereal box weights

A cereal company in Guadalajara sells 500g boxes. Box weights follow $X \sim N(500, 9)$ (variance 9, so $\sigma = 3$). Boxes weighing less than **496g** must be repackaged. [19]

- What is the probability that a box is repackaged?
- For 15,000 boxes, what is the approximate (normal) distribution of the number repackaged?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

```
1 n <- 15000
2
3 # (a) the probability that one box comes out under 496g
4 # (b) the mean and variance of the NUMBER of repackaged boxes out of 15,000
```

(c) Profits per box are P_1 if at least 496g and P_2 if less. For 15,000 boxes, what is the approximate distribution of total profits?

Total profit is $\Pi = (15000 - Y)P_1 + YP_2 = 15000 P_1 - Y(P_1 - P_2)$. With $a = P_1 - P_2$, a linear function of the (approximately normal) Y is normal:

$$\Pi \sim N(15000 P_1 - 1368.2 a, 1243.4 a^2)$$

Exam practice 5 — Online marketplace visitors

The number of visitors X to a Mexican online marketplace during one minute is $\text{Poisson}(\lambda = 150)$. [110]

- Thinking of X as the sum of 60 i.i.d. $\text{Poisson}(2.5)$ random variables, what is the asymptotic normal distribution associated with X ?
- Provide an approximate 90% probability interval for X using the asymptotic distribution.
- Calculate the exact probability (based on $\text{Poisson}(150)$) that X is within the interval from (b).

Exercise [↺ Start Over](#) [💡 Show Hint](#)

[▶ Run Code](#)

```
1 lambda <- 150
2
3 # (a) the mean and variance of X, seen as a SUM of 60 iid Poisson(2.5)
4 # (b) an interval around that mean holding about 90% of the distribution
5 # (c) the exact Poisson probability of landing inside that interval
```

Bonus: why is $S = \sum_{i=1}^n X_i$ an unbiased estimator of $n\mu_X$?

$E[S] = E[\sum_{i=1}^n X_i] = \sum_{i=1}^n E[X_i] = n\mu_X$ — the expected value of S equals the parameter, which is exactly the definition of unbiasedness.



Exam practice 6 — Franchise revenues

You own **60 taquería** franchises and **30 torta** franchises. Monthly revenue (thousands of pesos) for each franchise is i.i.d. with: [14]

- Taquerías: $\mu = 22, \sigma = 5$
- Tortas: $\mu = 17, \sigma = 4$

- a. What is the approximate distribution of the sample average of revenues for the 30 torta franchises?
- b. What is the approximate distribution of the sample average of revenues across all 90 franchises?

Exercise

↺ Start Over

💡 Show Hint

▶ Run Code

- 1 # (a) the mean and variance of the average revenue of the 30 torta franchises
- 2 # (b) the mean and variance of the average revenue across all 90 franchises